

STA623 - Bayesian Data Analysis

Practical 5: the Bayesian workflow: regression, model checking, comparison and hierarchical models

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Goals. Carry out a complete Bayesian analysis: prior predictive checks, fitting, diagnostics, interpretable summaries, posterior predictive checks, model comparison with LOO, and hierarchical modelling with partial pooling. Accompanies Lecture 5.

The helper `loo_from_ll()` from the lecture notes is defined in the (hidden) setup chunk.

Exercise 1: logistic regression, from priors to predictions

A cross-sectional study of 400 adults attending an outpatient clinic records hypertension (yes/no), age and sex:

```
set.seed(51)
N <- 400
bp <- tibble(age = round(runif(N, 18, 75)), male = rbinom(N, 1, 0.45)) |>
  mutate(hyp = rbinom(N, 1, plogis(-1.2 + 0.6 * (age - 45) / 10 + 0.3 * male)),
         age10_c = (age - 45) / 10)
```

Model: $\text{logit } P(\text{hyp}_i = 1) = \beta_0 + \beta_1 \text{age10_c}_i + \beta_2 \text{male}_i + \beta_3 \text{age10_c}_i \times \text{male}_i$.

- Prior predictive check.** Compare two prior sets: (A) all coefficients $\sim \mathcal{N}(0, 10^2)$; (B) $\beta_0 \sim \mathcal{N}(0, 1.5^2)$, $\beta_1, \beta_2 \sim \mathcal{N}(0, 1)$, $\beta_3 \sim \mathcal{N}(0, 0.5^2)$. For each, simulate the prior predictive distribution of the *prevalence of hypertension among 70-year-old men*. Which prior set is more sensible?
- Fit the model with prior set B in NIMBLE (4 chains). Check convergence.
- Report posterior medians and 95% credible intervals for the odds ratio per 10 years of age in women, the odds ratio per 10 years in men, and the male-vs-female odds ratio at age 45. What is the posterior probability that the effect of age is stronger in men than in women?
- Plot predicted prevalence by age for men and women with 95% credible bands.

Exercise 2: posterior predictive checks and LOO for count data

Number of outpatient visits in one year for 250 people living with diabetes, by whether they live more than 10 km from the clinic:

```
set.seed(52)
M <- 250
visits_df <- tibble(far = rbinom(M, 1, 0.4)) |>
  mutate(visits = rnbinom(M, size = 2, mu = exp(1.8 - 0.4 * far)))
visits_df |> group_by(far) |> summarise(n = n(), mean = mean(visits), var =
  ↪ var(visits))
```

```
# A tibble: 2 x 4
  far      n mean  var
<int> <int> <dbl> <dbl>
1     0  143  5.53 22.2
2     1  107  3.93 11.0
```

- (a) Fit a Poisson regression and a negative binomial regression ($\log \mu_i = \alpha_0 + \alpha_1 \text{far}_i$; use $\alpha_0 \sim \mathcal{N}(1, 1.5^2)$, $\alpha_1 \sim \mathcal{N}(0, 1)$, and for the negative binomial a dispersion $\phi \sim \text{Exp}(1)$ with size $r = 1/\phi$).
- (b) Compare the posterior of the rate ratio e^{α_1} under the two models.
- (c) Posterior predictive check: for each model simulate 1,000 replicated datasets (each from a different posterior draw). Use as test statistics the standard deviation of visits, the proportion of people with 0 or 1 visits, and the difference in mean visits between groups. Compute posterior predictive p-values.
- (d) Compare the models with LOO. Which would you report and why?

Exercise 3: a hierarchical model for multi-site data

HIV testing uptake among antenatal clinic attendees at 12 health centres (numbers offered a test and numbers accepting):

```
hc <- tibble(centre = sprintf("HC%02d", 1:12),
  offered = c(220, 35, 140, 12, 310, 60, 88, 25, 190, 8, 150, 45),
  accepted = c(198, 26, 131, 12, 270, 55, 71, 22, 176, 5, 139, 37))
hc |> mutate(raw = accepted / offered)
```

```
# A tibble: 12 x 4
  centre offered accepted  raw
  <chr>    <dbl>    <dbl> <dbl>
1 HC01      220      198  0.9
2 HC02       35       26 0.743
```

3	HC03	140	131	0.936
4	HC04	12	12	1
5	HC05	310	270	0.871
6	HC06	60	55	0.917
7	HC07	88	71	0.807
8	HC08	25	22	0.88
9	HC09	190	176	0.926
10	HC10	8	5	0.625
11	HC11	150	139	0.927
12	HC12	45	37	0.822

- (a) Fit three models for the centre-specific acceptance probabilities p_j : complete pooling (one common p), no pooling (independent $p_j \sim \text{Beta}(1, 1)$; you may use conjugacy rather than MCMC), and partial pooling with $\text{logit}(p_j) = \mu + \sigma z_j$, $z_j \sim \mathcal{N}(0, 1)$, $\mu \sim \mathcal{N}(1.5, 1.5^2)$, $\sigma \sim \text{Exp}(1)$.
- (b) Plot the three sets of estimates (with 95% intervals for no and partial pooling) against the centre sample size. Which centres are shrunk the most?
- (c) The district target is 85% uptake. For each centre, compute $P(p_j < 0.85 \mid \text{data})$ under no pooling and under partial pooling. Which centres would you flag for supportive supervision under each model? Comment on HC04 and HC10.
- (d) Refit the hierarchical model with the *centred* parameterisation, $\text{logit}(p_j) \sim \mathcal{N}(\mu, \sigma^2)$. Compare $\widehat{R}$ and ESS for σ between the two parameterisations.